

Predicting Hit Songs Using Repeated Chorus

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Question and evidence

Can a 15-second chorus distinguish a source year-end Billboard hit (class 1) from another chart song (class 0)? Both classes can have charted. The public feature-table task differs from the supplied reference paper. This update preserves its labels and original historical-test membership.

The audited table contains 751 songs, 71 provided artist names and 518 raw audio features. Class counts are 385 other-chart songs and 366 year-end hits. Original recordings are unavailable. Names are grouping metadata, never model inputs.

Method

The original development partition has 597 songs from 52 artist-name groups. We compare 35 predeclared settings with fixed seeds. 5 outer folds each use 3 inner folds for nested selection. The full development partition then selects the final candidate using inner-fold balanced accuracy. Historical labels never tune it.

The model accepts 518 ordered raw audio features. Selection criterion: 3-fold development balanced accuracy; predeclared ID tie break.

Measured outcome

Run **v2_nested_001**. Final development candidate: **lr_mi_50**. lr_mi_50 records 52.9% balanced accuracy for nested development. Full-development mean tuning balanced accuracy: 52.9%. Nested predictions assess the selection procedure, rather than the single final refit.

Metric	Nested Development	Strict >75%
Accuracy	52.9%	Fail
Balanced Accuracy	52.9%	Fail
Precision	51.6%	Fail
Recall	50.0%	Fail
F1	50.8%	Fail

The approximate 95% group-bootstrap interval (48.3%-57.3%) includes 50%; a reliable advantage over chance is not established.

The five-metric target is not met in this evaluation. Fresh-test confirmation is unavailable.

Historical comparison and limits

The frozen candidate scored 55.2% balanced accuracy on the same 154 historical songs from 19 groups. The paired change from V1 is +8.5 percentage points, with a group-bootstrap interval of -1.6 to +16.9 points. This interval includes zero. These already known records provide a historical comparison, not fresh confirmation.

The baseline demo remains active. New audio predictions are exploratory. No fresh test or real-song embedding results exist. Full data provenance, trial records and the exact software checks accompany the project.

[1] Eric Liu, CS229, 2021: [reference report](#).
[2] [Antonios Malak feature dataset](#), pinned commit 838e76f, Apache-2.0.
[3] [Scikit-learn nested cross-validation](#). Local run manifests and predictions supply all reported metrics.